

Longformer with MBTI-Augmented Big Five Classification

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Longformer-Based Personality Trait Modeling with MBTI-Augmented Big Five Classification

1. Introduction

The Big Five (OCEAN) and MBTI are widely used personality frameworks. While differing in structure, they capture complementary aspects of personality and are often used in theory-driven NLP research (Myers, 1987).

2. Motivation and Problem Statement

Transformer models have advanced personality trait classification, but most (e.g., BERT) are limited to 512 tokens—insufficient for long-form texts like those in the PANDORA dataset. Since Big Five traits rely on subtle, distributed cues, capturing long-range context is essential. Few studies have explored models like **Longformer**, which extends the context window to 4096 tokens using sparse attention.

3. Objective and Contributions

This paper proposes a two-stage system: (1) predict MBTI traits using classical ML models trained on Big Five traits, and (2) use predicted MBTI vectors in a Longformer-based Big Five classifier. Contributions include early fusion of MBTI embeddings, handling long-form input, and benchmarking against recent state-of-the-art methods.

The working assumption was that MBTI-augmented approach enhances classification accuracy for all five personality traits. My contributions include:

- Implementing early fusion of MBTI embeddings into the trait prediction pipeline.
- Leveraging Longformer for full-context personality classification.
- Using multi-class classification per trait (low/medium/high) with trait-wise loss balancing.
- Studying the effectiveness of the proposal against other recent state-of-the-art studies based on traditional machine learning techniques and deep learning architectures.

4. Literature Review

- Personality Trait Detection via Transfer Learning (Bashar Alshouha et al.)
- PANDORA Talks: Personality and Demographics on Reddit (Matej Gjurković et al.)
- Continuous Output Personality Detection Models via Mixed Strategy Training (Rong Wang)
- Myers & McCaulley (1987) – foundational work on the MBTI framework.

5. Dataset and Preprocessing

The PANDORA dataset is a comprehensive compilation of Reddit comments from over 10,000 users, labeled with three personality models (Big Five, MBTI, Enneagram) and demographic data (age, gender, location) (Gjurković et al., 2021).

Important to note that the topics are not tied to the use case in the evaluation stage (job interviews). Furthermore, we use Author Profiles dataset that is populated with author MBTI profiles, however with a nuance of missing larger part of input.

Data Augmentation

To address trait imbalance, we chunked user data (max 10 comments each), merged known MBTI profiles, and trained classical ML models to predict missing MBTI values from binned Big Five traits (quantile-based). We then applied stratified sampling to create a balanced 10,000-row dataset. Text preprocessing included lowercasing and removal of URLs, Markdown links, and user mentions.

6. Methodology & Architecture

Stage 1 — MBTI Prediction Model

- Model: Logist Regression (for ei, ft, jp), class_weight="balanced"
- XGBClassifier (for ns) with eval_metric="logloss"
- Input: Binned Big Five personality traits (low, medium, high → encoded as integers)
- Output: 4 binary MBTI axes: ei, ns, ft, jp
- Preprocessing:
 - Sparse MBTI-labeled data split into known and missing subsets per axis
 - SMOTE used to oversample the minority class and balance training data
- Outputs: Predicted MBTI vectors for unlabeled instances, used for MBTI-augmented trait classification in Stage 2

Stage 2 — MBTI-Augmented Big Five Classification

- We used early fusion by concatenating the predicted MBTI vector with the [CLS] embedding. The output layer predicts 5 traits × 3 classes using cross-entropy loss, with trait-specific class weights to address imbalance.
- Output: A 15-dimensional vector representing predictions for 5 personality traits,

7. Results: Performance Overview

The MBTI-augmented Longformer model achieved moderate performance, with validation F1 at 0.51 and test F1 at 0.27. Extraversion was the strongest trait, while others like Conscientiousness and Agreeableness underperformed. Despite leveraging MBTI inputs and extended context, results remained below benchmarks like M3 (RoBERTa-based).

Observations

- Longformer's large context window (up to 4096 tokens) enabled modeling long text sequences, but this advantage did not translate into significantly higher performance.

- The model was computationally expensive to train, requiring significant GPU resources (A100-class hardware) and small batch sizes (2–4) due to memory constraints.
- MBTI fusion showed promise in validation but did not generalize well to unseen, real-life data, suggesting possible overfitting or limitations in the fusion strategy.

Benchmarking Against RoBERTa Models and Golden Standard

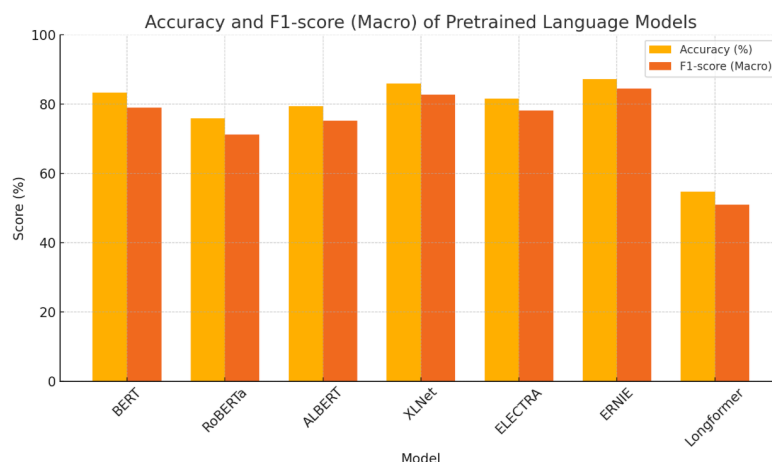
M1 and M3 are RoBERTa-based models from Wang (2023). M1 is direct tuning on RoBERTa with manually set hyperparameters, while M3 uses fine-tuning, parameter selection, and mixed precision training. Longformer outperformed M1 and M3 only on Extraversion; for other traits, it underperformed, especially on Neuroticism.

Big Five Trait	Accuracy (M1)	F1 Score (M1)	Accuracy (M3)	F1 Score (M3)	Accuracy (Longformer - Validation)	F1 Score (Longformer - Validation)
Agreeableness	0.59	0.45	0.79	0.78	0.51	0.5
Conscientiousness	0.75	0.64	0.88	0.87	0.58	0.57
Extraversion	0.6	0.44	0.61	0.47	0.69	0.55
Neuroticism	0.51	0.34	0.62	0.48	0.37	0.38
Openness	0.75	0.66	0.84	0.82	0.56	0.54

Source: Adapted from Wang (2023)

Golden Standard

Longformer shows significantly lower performance compared to benchmark models, with 54.7% accuracy and 51.0 F1, versus over 80% accuracy and 75–84 F1 for models like ERNIE and XLNet. This highlights a gap despite Longformer's extended context capabilities.



Source: Adapted from Alshouha (Bashar Alshouha et al.)

Validation vs Real-Life Evaluation

The model's performance drops notably from validation to real-life data across all traits. While Extraversion and Openness retain some predictive strength, traits like Conscientiousness and Agreeableness show steep declines, with real-life F1 scores as low as 0.21. Overall, the macro F1 drops from 0.51 to 0.27, indicating limited generalization and highlighting challenges in applying the model to real-world scenarios.

Trait	Accuracy (Validation)	F1 Score (Validation)	Accuracy (Real-Life)	F1 Score (Real-Life)
Openness	0.57	0.55	0.47	0.32
Conscientiousness	0.59	0.57	0.31	0.24
Extraversion	0.69	0.55	0.47	0.37
Agreeableness	0.51	0.51	0.19	0.21
Neuroticism	0.38	0.38	0.41	0.19
Macro Average	0.55	0.51	0.37	0.27

8. Limitations and Future Work

The model's performance is limited by the scarcity of MBTI labels, its two-stage complexity, and poor generalization to real-world data. Future work may explore latent embeddings, prompt-based inference, and applications like job fit prediction or contrastive learning between MBTI and Big Five traits.

9. Conclusion

Integrating MBTI predictions into a Longformer-based Big Five classifier offers a promising direction for modeling personality from long-form text. While the model shows strength in handling extended input, its real-world performance remains limited, suggesting the need for **stronger fusion techniques** and **further optimization**.

References

- *Manual: A Guide to the Development and Use of the Myers-Briggs Type Indicator* (Myers & McCaulley, 1987)
- Gjurković, M., et al. (2021). *PANDORA Talks: Personality and Demographics on Reddit*.

- Wang, R. (2023). *Continuous Output Personality Detection Models via Mixed Strategy Training*.
- Alshouha, B., et al. (2022). *Personality Trait Detection via Transfer Learning*.